FEDERATED UNLEARNING THESIS

Month 2-3 Progress Report

Gate 2 closure (hand-written FedAvg) and Month 3 Week 9 (IID vs. Non-IID)

Generated 2026-09-10 from saved metrics.json files under results/. No number below is hand-typed.

Gate 2 — FedAvg from Scratch (closed 2026-09-07)

Federated Averaging was implemented by hand and verified against 13 synthetic tests before being run on real MNIST data. The canonical protocol uses 5 IID clients, full participation, one local epoch per round, batch size 128, and five communication rounds, matched against a centralized SGD baseline trained on the same 51,000-example split from the same initial weights.

Canonical result and controlled variants

Setting	Centralized acc.	FedAvg acc.	FedAvg F1	Gap (pp)
Canonical (K=5, E=1)	94.76%	90.99%	90.86%	-3.77
K=10 (double clients)	94.76%	89.55%	89.37%	-5.21
E=2 (double local epochs)	96.17%	92.31%	92.21%	-3.86

Every predicted number the student wrote before either variant ran (examples per client, optimizer-step totals, exposures, communication bytes) matched the executed result exactly. Doubling client count widened FedAvg's gap to centralized at double the communication cost; doubling local epochs raised both methods' absolute accuracy without changing communication or the relative gap. Reviewed and passed 2026-09-07.

Month 3, Week 9 — IID vs. Non-IID FedAvg

With Gate 2 closed, Month 3 asks what happens to FedAvg once clients stop being interchangeable random samples. A pathological shard-based Non-IID partition (McMahan et al. 2017, Section 3: sort by label, two shards per client) was compared against the existing IID partition, training from identical initial weights under the same K=5, C=1, E=1, B=128, R=5 protocol as the Gate 2 canonical run.

Partition	Test accuracy	Test macro F1
IID	90.99%	90.86%
Non-IID (2 shards/client)	64.01%	57.80%

The IID number exactly reproduces Gate 2's canonical FedAvg result — a built-in sanity check that this new runner is wired correctly, not a new measurement. The Non-IID run drops 26.98 percentage points. Each Non-IID client's saved class histogram is dominated by one or two digits, while every IID client stays close to flat across all ten — the heterogeneity is visible in the saved data, not just asserted. This is one Non-IID severity, not a Dirichlet sweep, and not a FedProx comparison; those are Weeks 10 and 11.

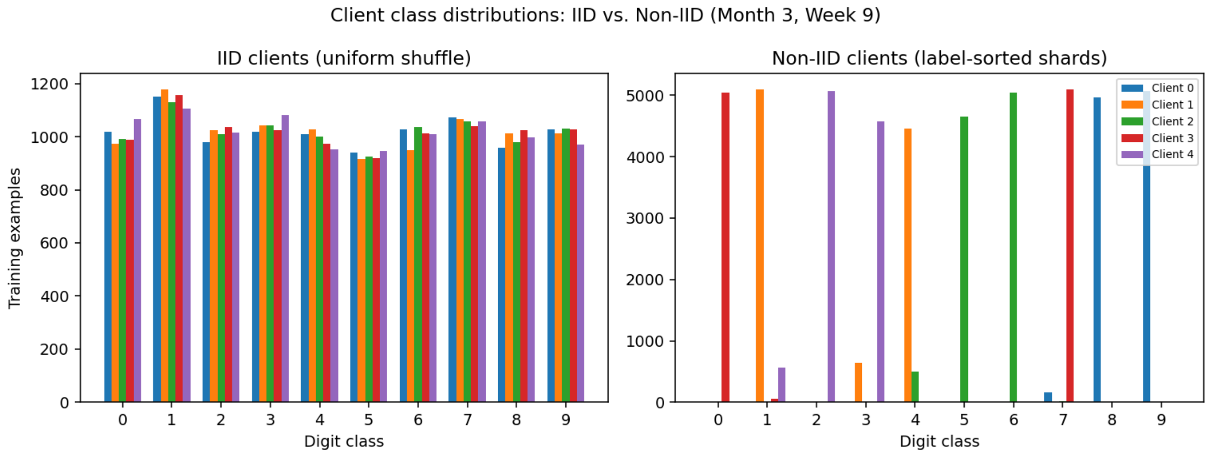

Figure: per-client training-set class distribution, IID (left) vs. Non-IID (right).

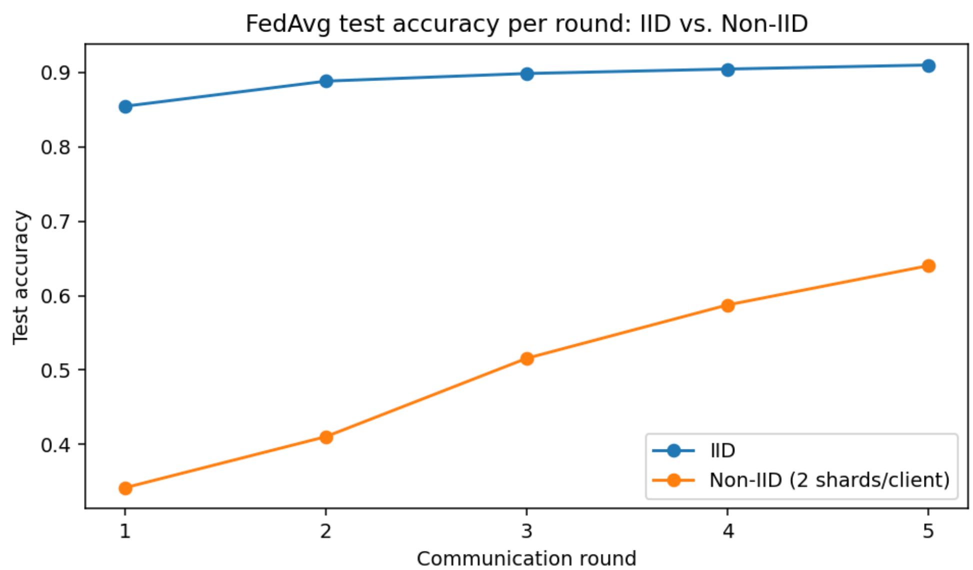

Figure: FedAvg test accuracy per communication round, IID vs. Non-IID.

Month 3, Week 10 — Dirichlet(α) Severity Gradient

Week 9 showed one pathological extreme. Week 10 asks how FedAvg degrades as heterogeneity is dialed continuously, using Hsu, Qi & Brown (2019)'s per-class Dirichlet(α) label-skew scheme: a smaller α concentrates each digit class onto fewer clients; a larger α keeps classes close to uniformly spread. Three configs (α = 1.0, 0.5, 0.1) each retrain centralized SGD and hand-written FedAvg from identical initial weights under the same protocol as Week 8/9.

Setting	Centralized	FedAvg	Gap (pp)
IID (Week 8/9)	94.76%	90.99%	-3.77
Dirichlet α=1.0	94.76%	90.59%	-4.17
Dirichlet α=0.5	94.76%	89.83%	-4.93
Dirichlet α=0.1	94.76%	71.60%	-23.16
Pathological shards (Week 9)	94.76%	64.01%	-30.75

The severity gradient is nonlinear in α, matching Hsu et al.'s own finding: α=1.0 and α=0.5 both stay within about 1.4 points of the IID result, while α=0.1 causes a much larger drop. Dirichlet skew also unevenly skews client *quantity*, not just label mix — training-set sizes ranged from about 6,100 to 16,700 examples across the 5 clients at every α tested, a real, literature-typical side effect of the per-class draw. Placing Week 9's pathological-shard result in the same table shows “Non-IID” is a spectrum: Dirichlet α=0.1 is meaningfully less severe than the 2-shard scheme, even though both are informally called “severe.”

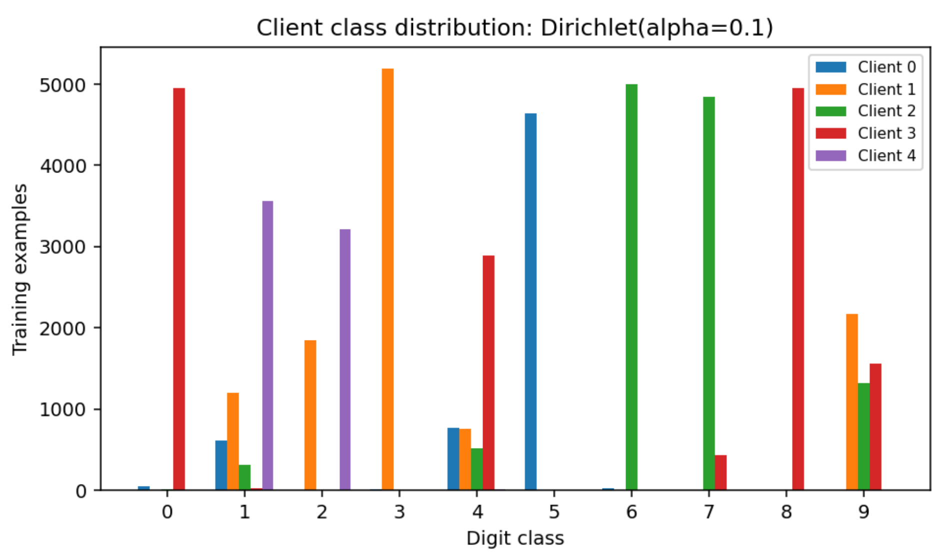

Figure: per-client training-set class distribution at Dirichlet α=0.1.

Month 3, Week 11 — Does FedProx Help?

Li et al. (2020)'s FedProx adds a proximal term (mu/2)*||w - w_global||² to each client's local loss, specifically to limit the client drift Weeks 9-10 measured. mu=0 is verified by test to reproduce hand-written FedAvg exactly (client weights and a full server round), so FedProx here is a strict generalization, not a parallel reimplementation. The experiment reuses Week 10's exact Dirichlet(α=0.1) partition, seeds, and initial weights, sweeping mu ∈ {0.01, 0.1, 1.0} at two local-epoch counts.

Method	E=1 accuracy	E=5 accuracy
FedAvg (mu=0)	71.60%	76.28%
FedProx (mu=0.01)	71.43%	75.33%
FedProx (mu=0.1)	69.65%	69.65%
FedProx (mu=1.0)	60.15%	57.30%

FedProx did not beat FedAvg at any swept mu, in either local-epoch setting — reported plainly, per the plan's own rule that a negative result must remain reportable. This does not contradict Li et al.'s paper: their clearest reported gains come under systems heterogeneity (clients doing genuinely different amounts of local work) and longer training horizons, neither of which this 5-round, uniform-local-epoch experiment tests. Three checks rule out an implementation bug: mu=0 equivalence is tested, not assumed; the E=1 and E=5 round-by-round histories genuinely differ; and one striking coincidence — mu=0.1 landing on the identical 69.65% at both E=1 and E=5 — was traced to the underlying per-round trajectories and confirmed real, not a copy-paste error. Full discussion: reports/month3_week11_fedprox.md.

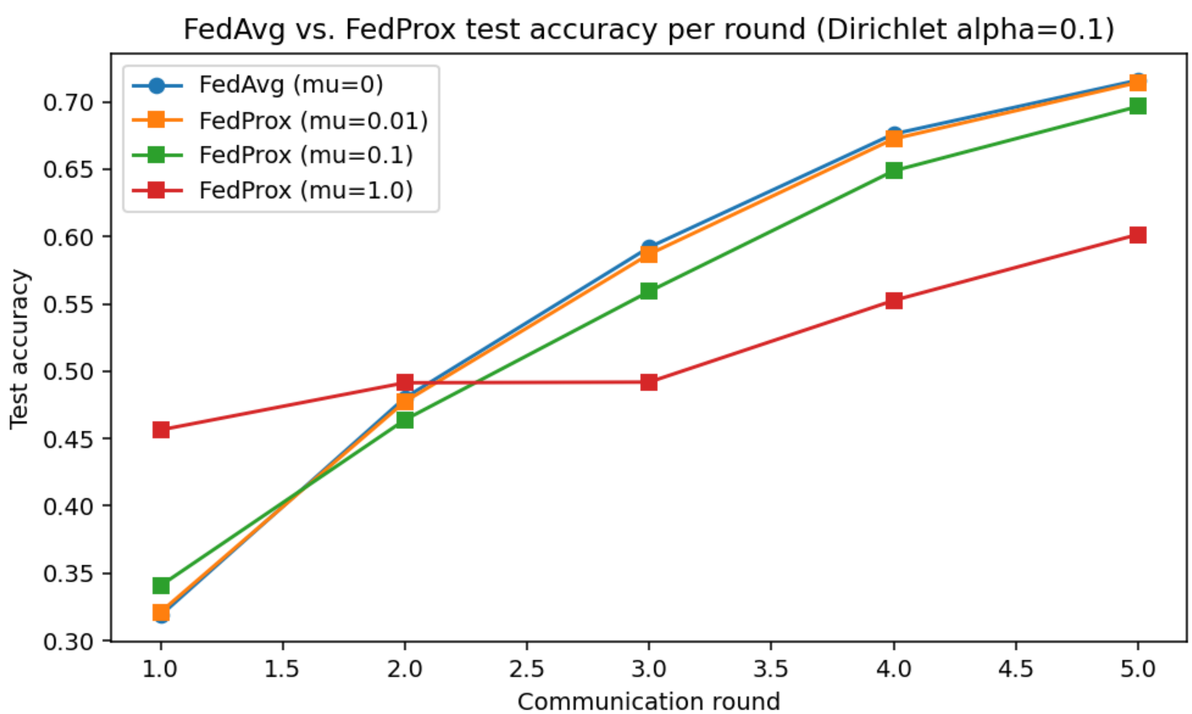

Figure: FedAvg vs. FedProx test accuracy per round at Dirichlet α=0.1, E=1.

Month 3, Week 12 — The Gate 3 Benchmark

The complete FedAvg-vs-FedProx table (plan §9) across all four data distributions Weeks 9-10 characterized. FedAvg legs reuse already-verified evidence from Weeks 8-10; only FedProx (fixed mu=0.01, the least-detrimental value Week 11 found) was run fresh at IID and alpha=1.0/0.5, with alpha=0.1 reused directly from Week 11 rather than rerun.

Setting	FedAvg	FedProx (mu=0.01)
IID	90.99%	90.88%
Dirichlet α=1.0	90.59%	90.54%
Dirichlet α=0.5	89.83%	89.72%
Dirichlet α=0.1	71.60%	71.43%

FedProx is very slightly below FedAvg at every setting, including IID — consistent with Week 11's finding at a wider mu sweep, now shown across the full heterogeneity range at one fixed mu. Communication is identical across every row (20,354,000 bytes) since it depends only on model size, client count, and rounds, not which algorithm is used. Full discussion, including why this does not contradict Li et al. (2020): reports/month3_week12_benchmark.md.

Gate 3's learning check (reports/gate3_self_check.md) asks eight questions on statistical vs. systems heterogeneity, client drift mechanics, and correctly scoping FedProx's negative result — prepared, not yet answered. Gate 3 remains open until it is reviewed; Federated Unlearning remains blocked until Gates 3 and 4 both close.